EEE3096S Practical 3 — Voltage Mirror

Student report template

Names/student numbers: _____

Date: _____ Build configuration: _____

Assessment: live demonstration 80 marks; report 20 marks. Enter your own measurements. Suggested input levels are setup targets, not results. Submit the four completed functions with this report.

1. Measurements and predictions (8 report marks)

Measured analogue reference V_R: 3.2946 V

DMM range/resolution: 10 V DC range, 5 significant figures (0.1 mV resolution)

Nominal ADC/DAC interval $V_R/4096$: $3.2946/4096 = 0.8043$ mV

Predicted coarse output step $V_R/64$: $3.2946/64 = 51.48$ mV

Measure PA5 and PA4 independently with the DMM. Record the actual LCD codes. Calculate the nominal DAC output $DV_R/4096$ from each recorded command.

Input target	Mode	Measured PA5 (V)	C	D	Predicted PA4 (V)	Measured PA4 (V)
∼0.800 V	FULL12					
∼0.800 V	COARSE6					
∼1.650 V	FULL12					
∼1.650 V	COARSE6					
∼2.500 V	FULL12					
∼2.500 V	COARSE6					

Generator: _____ Hz; _____ Vpp; _____ V offset

Verified PA5 minimum/maximum: _____ / _____ V

Measured adjacent coarse step: _____ mV

Measured minus predicted step: _____ mV

Percentage difference $100(\Delta V_{\text{measured}} - \Delta V_{\text{predicted}})/\Delta V_{\text{predicted}}$: _____ %

2. Two real scope captures (6 report marks)

Label PA5 input and PA4 output, modes, voltage scales and time scales. Use your own captures; simulated illustrations are not experimental evidence.

FULL12: input and output over approximately one triangle cycle

Replace with your own oscilloscope capture.
Save it as `full12.png`.

Capture 1 — FULL12. Show both traces over approximately one cycle. The small individual 12-bit steps need not be resolved.

COARSE6: adjacent settled plateaus and voltage cursors

Replace with your own oscilloscope capture.
Save it as `coarse6.png`.

Capture 2 — COARSE6. Show both channels, adjacent settled output plateaus and the cursor measurement used for your step comparison.

3. Explanation (6 report marks; at most one page)

Replace the prompts and blank spaces with concise answers supported by your measurements. Keep the whole written explanation on this page.

Mapping. State your FULL12 and COARSE6 relationships. How many distinct coarse commands exist, what is their spacing, and why is the maximum 4032?

Measured steps. Compare the measured coarse step with $V_R/64$. Use your readings and instrument limits to discuss any difference.

Resolution and accuracy. Explain why a smooth FULL12 trace does not establish continuous voltage resolution, and why 12-bit resolution does not guarantee voltage accuracy. Both hardware converters stay at 12 bits.

One DC result. Compare $CV_R/4096$ with measured input, and $DV_R/4096$ with measured output. Explain why copying the code does not guarantee identical physical voltages.

Live demonstration: prepare all three DC levels; show one safe tutor-selected input and both triangle modes. Explain the actual codes, $V_R/64$ step prediction, ADC/DAC register fields, bounded wait and W1C flags.